BTVNANOCOMMISSIONING · ZB_ZC_RATIO WORKFLOW

# Z(→ℓℓ) + b/c AK4 jets in Run 3

Summer22 DY→ℓℓ MC with AK4 jets: samples, cutflow, kinematics, and a comparison with the boosted AK8 selection

11 October 2026

# Campaign and setup

Goal: measure σ(Z+b) / σ(Z+c) with an AK4 jet recoiling against Z→ee/μμ (switched from the boosted AK8 jet after the group meeting). First step: tune the selection on DY→ℓℓ MC.

**MC CAMPAIGN**

## Summer22

Run3Summer22 = 2022 eras C–D (pre-EE)

**FORMAT**

## NanoAODv12

130X MC conditions

**ENERGY (√s)**

## 13.6 TeV

Run 3 pp collisions

**NORMALIZATION**

## 7.98 $fb^{-1}$

2022 C–D golden JSON; brilcalc check pending

MC only, no data yet. Processed: 1 file per sample (test run), the same files as the AK8 test. Code: BTVNanoCommissioning, branch coffea_machine, workflow Zb_Zc_ratio (AK4, commit f8748a6); the AK8 selection runs as Zb_Zc_ratio_AK8.

# MC samples: DY→ℓℓ, stitched in pT(ℓℓ)

```
/DYto2L-2Jets_MLL-50[_PTLL-X]_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/
  Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-vN/NANOAODSIM
```

| Sample | LHE pT(ℓℓ) used | σ [pb] | DAS files | DAS events | Test: events used |
|---|---|---|---|---|---|
| Inclusive (MLL-50) | < 100 GeV | 6346 | 525 | 93.8 M | 197,672 |
| PTLL-100 | 100–200 GeV | 105.0 | 418 | 56.4 M | 161,921 |
| PTLL-200 | 200–400 GeV | 11.24 | 335 | 44.7 M | 91,186 |
| PTLL-400 | 400–600 GeV | 0.643 | 424 | 41.6 M | 2,727 |
| PTLL-600 | > 600 GeV | 0.0994 | 185 | 9.50 M | 939 |

PTLL-X samples are inclusive above X, so each is kept to its own LHE pT(ℓℓ) slice. σ(PTLL-X) = 6346 pb × fraction of inclusive events with pT(ℓℓ) > X (measured on 27.7 M inclusive events). "Test: events used" = events of 1 file inside the slice. With no boosted-jet requirement the inclusive sample (pT(ℓℓ) < 100 GeV) gives 87% (Z→ee) / 89% (Z→μμ) of the scaled yield from only 11–13% of the raw selected events, so its large per-event weights limit the MC statistics.

# Event selection, per channel

## Z→ee / Z→μμ (unchanged)

- Trigger: HLT_Ele30_WPTight_Gsf / HLT_IsoMu24
- Leptons pT > 25 GeV, |η| < 2.4; leading pT > 35 GeV
- e: tight cut-based ID, IP cuts, EB–EE gap veto
- μ: tight ID, PF isolation < 0.15
- Opposite sign (OS)
- 71 < m$\ell\ell$ < 111 GeV
- PuppiMET < 50 GeV

## Leading AK4 jet ("jet" step)

- AK4 PUPPI jets, ΔR > 0.4 from selected leptons
- Tight-lepton-veto jet ID (with NanoAODv12 jetId fix)
- pT > 30 GeV, |η| < 2.5; ≥ 1 such jet, leading one used
- No Z pT requirement

## Weights and flavour

- genWeight × pileup × lepton ID/iso/reco SFs
- No trigger SF yet
- Jet flavour from hadronFlavour: b (5), c (4), light; no bb/cc split for AK4
- Also labelled by |partonFlavour|: b, c, uds, g, none (pages 21+, CMS recommendation on p. 21)
- Tagger: PNet AK4 B, CvL, CvB

# Cutflow, Z→ee: raw MC events per step

| Sample | all | trigger | leptons | OS | Zmass | MET | jet | jet / MET |
|---|---|---|---|---|---|---|---|---|
| Inclusive | 197,672 | 32,660 | 9,802 | 9,745 | 9,396 | 8,872 | 2,347 | 26.5% |
| PTLL-100 | 161,921 | 46,448 | 13,156 | 13,037 | 12,032 | 10,642 | 10,214 | 96.0% |
| PTLL-200 | 91,186 | 31,202 | 11,594 | 11,431 | 10,398 | 8,584 | 8,466 | 98.6% |
| PTLL-400 | 2,727 | 1,013 | 458 | 447 | 396 | 287 | 286 | 99.7% |
| PTLL-600 | 939 | 229 | 101 | 100 | 82 | 54 | 54 | 100.0% |

**21,367** raw MC events pass the full Z→ee selection; the most (48%) come from PTLL-100. The AK4 jet step keeps 26% of inclusive (low-pT) Z events and ≥ 96% at pT($\ell\ell$) > 100 GeV.

Summer22 DY→$\ell\ell$ MC · 1 file per sample · AK4 jets · "all" = events in the sample's LHE pT($\ell\ell$) slice · 5

# Cutflow, Z→μμ: raw MC events per step

| Sample | all | trigger | leptons | OS | Zmass | MET | jet | jet / MET |
|---|---|---|---|---|---|---|---|---|
| Inclusive | 197,672 | 42,062 | 17,685 | 17,685 | 17,054 | 16,139 | 4,159 | 25.8% |
| PTLL-100 | 161,921 | 48,592 | 20,606 | 20,604 | 19,043 | 16,788 | 16,142 | 96.2% |
| PTLL-200 | 91,186 | 31,717 | 16,868 | 16,865 | 15,258 | 12,453 | 12,298 | 98.8% |
| PTLL-400 | 2,727 | 1,048 | 659 | 659 | 592 | 407 | 405 | 99.5% |
| PTLL-600 | 939 | 319 | 199 | 199 | 178 | 102 | 102 | 100.0% |

**33,106** raw MC events pass the full Z→μμ selection; the most (49%) come from PTLL-100. The AK4 jet step keeps 26% of inclusive (low-pT) Z events and ≥ 96% at pT($\ell\ell$) > 100 GeV.

Summer22 DY→$\ell\ell$ MC · 1 file per sample · AK4 jets · "all" = events in the sample's LHE pT($\ell\ell$) slice · 6

# Yields by AK4 jet flavour (7.98 fb$^{-1}$)

| Channel | light | c | b | total |
|---|---|---|---|---|
| Z→ee | 343,532.5 | 42,631.0 | 16,844.4 | 403,007.9 |
| fraction | 0.852 | 0.106 | 0.042 | |
| Z→μμ | 683,956.5 | 65,108.4 | 35,637.3 | 784,702.2 |
| fraction | 0.872 | 0.083 | 0.045 | |

### Still 1 file per sample

No negative yields now: 21,367 (Z→ee) and 33,106 (Z→μμ) raw events pass, vs ~1.4k / 2.1k with AK8. Single negative-weight amcatnlo events still leave jumps and empty bins in the tails of the plots. No b-tag requirement applied yet.

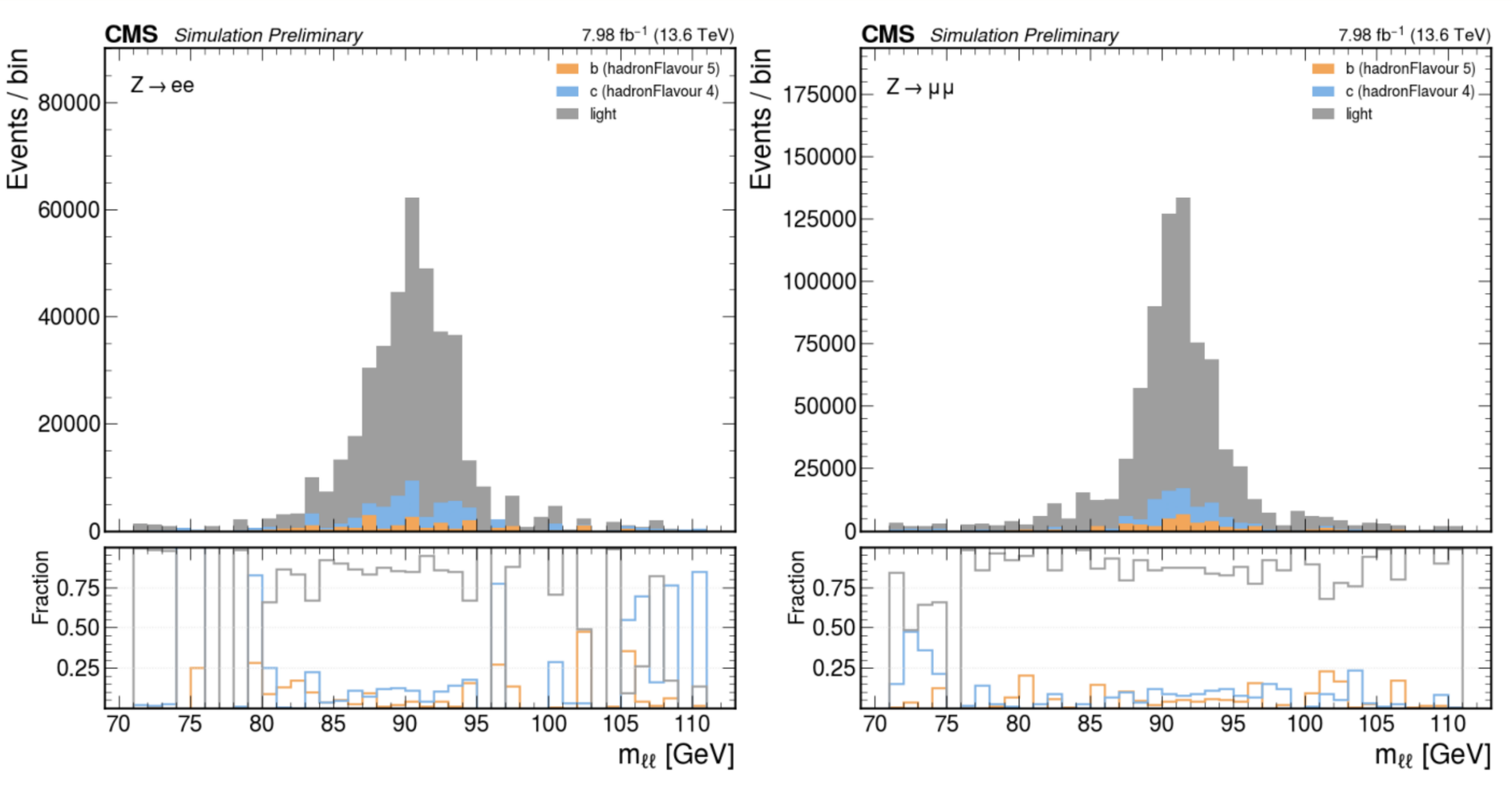
# AK8 vs AK4: same files, same Z selection

| | AK8 Z→ee | AK4 Z→ee | AK8 Z→μμ | AK4 Z→μμ |
|---|---|---|---|---|
| Raw events selected | 1,440 | 21,367 | 2,121 | 33,106 |
| Scaled yield | 2,259.7 | 403,007.9 | 2,784.7 | 784,702.2 |
| b fraction (b + bb) | 0.043 | 0.042 | −0.045 | 0.045 |
| c fraction (c + cc) | 0.118 | 0.106 | 0.244 | 0.083 |
| light fraction | 0.840 | 0.852 | 0.800 | 0.872 |
| Jet requirement | pT > 200, mSD > 40 | pT > 30 | pT > 200, mSD > 40 | pT > 30 |

AK8 flavour is from B/C hadron counts inside the jet (bb/b/cc/c); AK4 flavour is hadronFlavour. The AK4 selection keeps ~15× more events because it does not require a boosted Z. AK8 heavy-flavour fractions are unreliable at this statistics (negative Z→μμ b yield).

# Dilepton mass

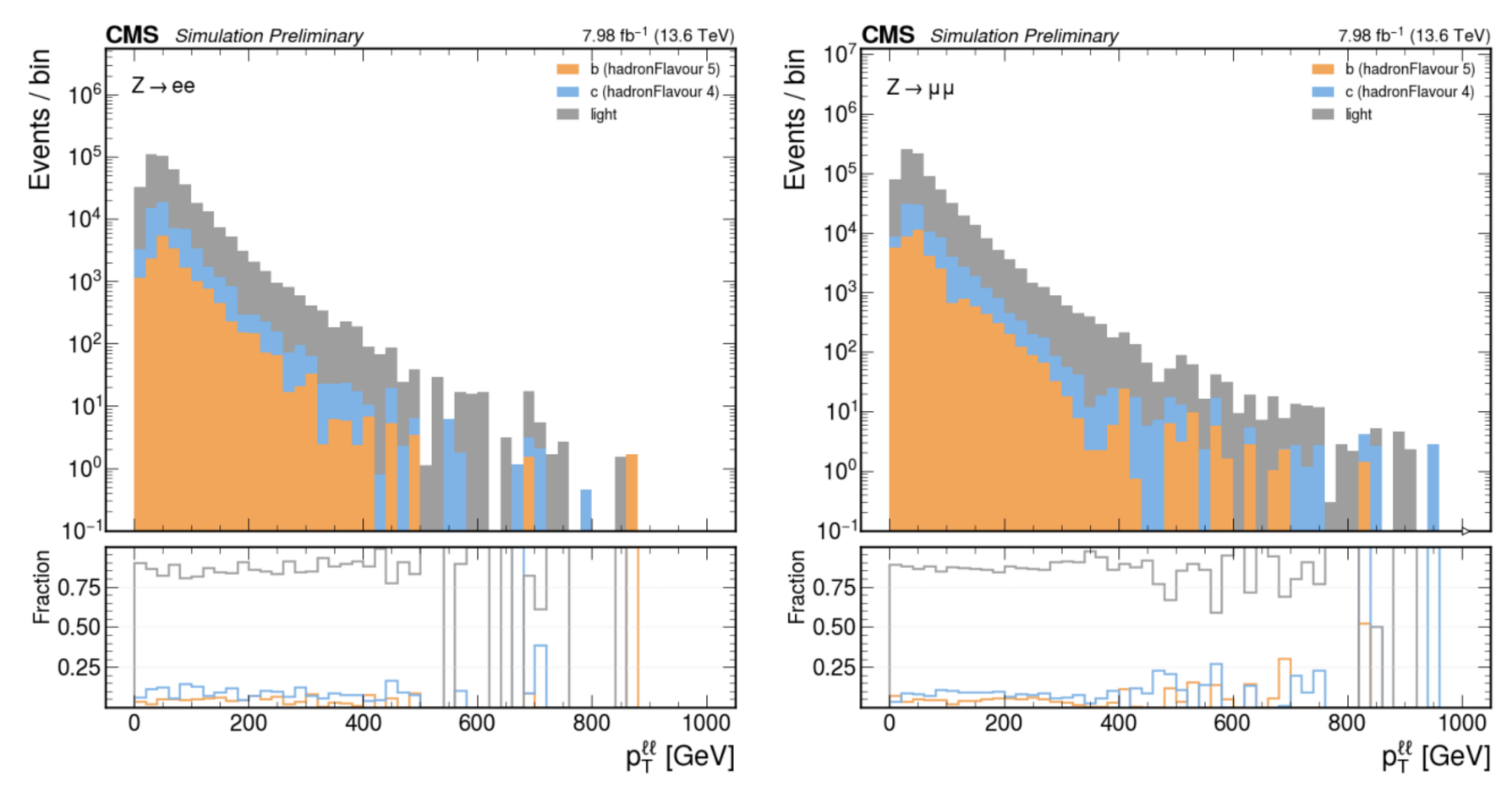
m(ℓℓ) after the full selection (71–111 GeV window): peaks at the Z mass in both channels. Linear y-axis.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 9

# Dilepton pT

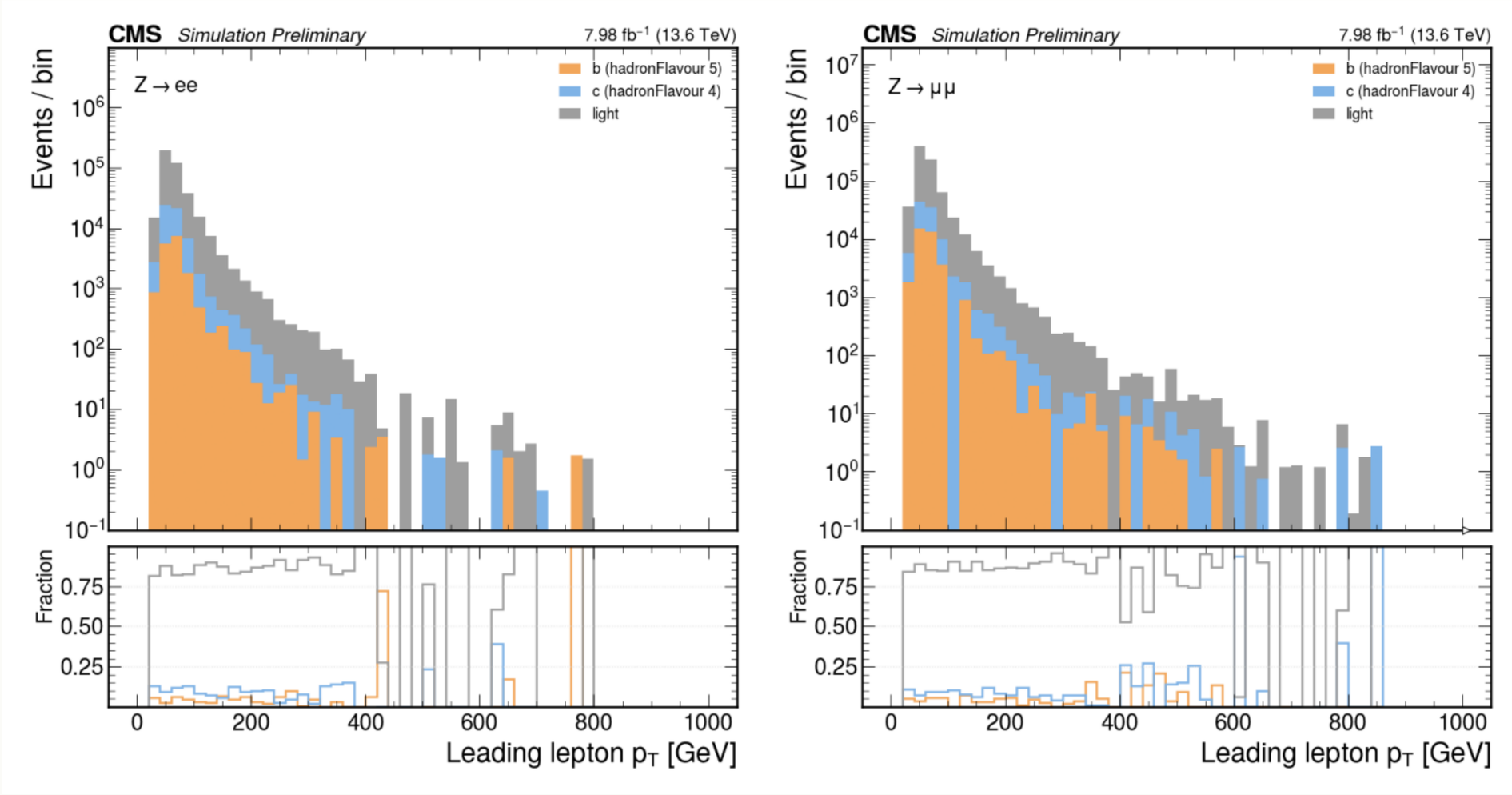
pT(ℓℓ) of the Z candidate after the full selection. No Z pT cut: dominated by low pT, unlike the boosted AK8 selection.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 10

# Leading-lepton pT

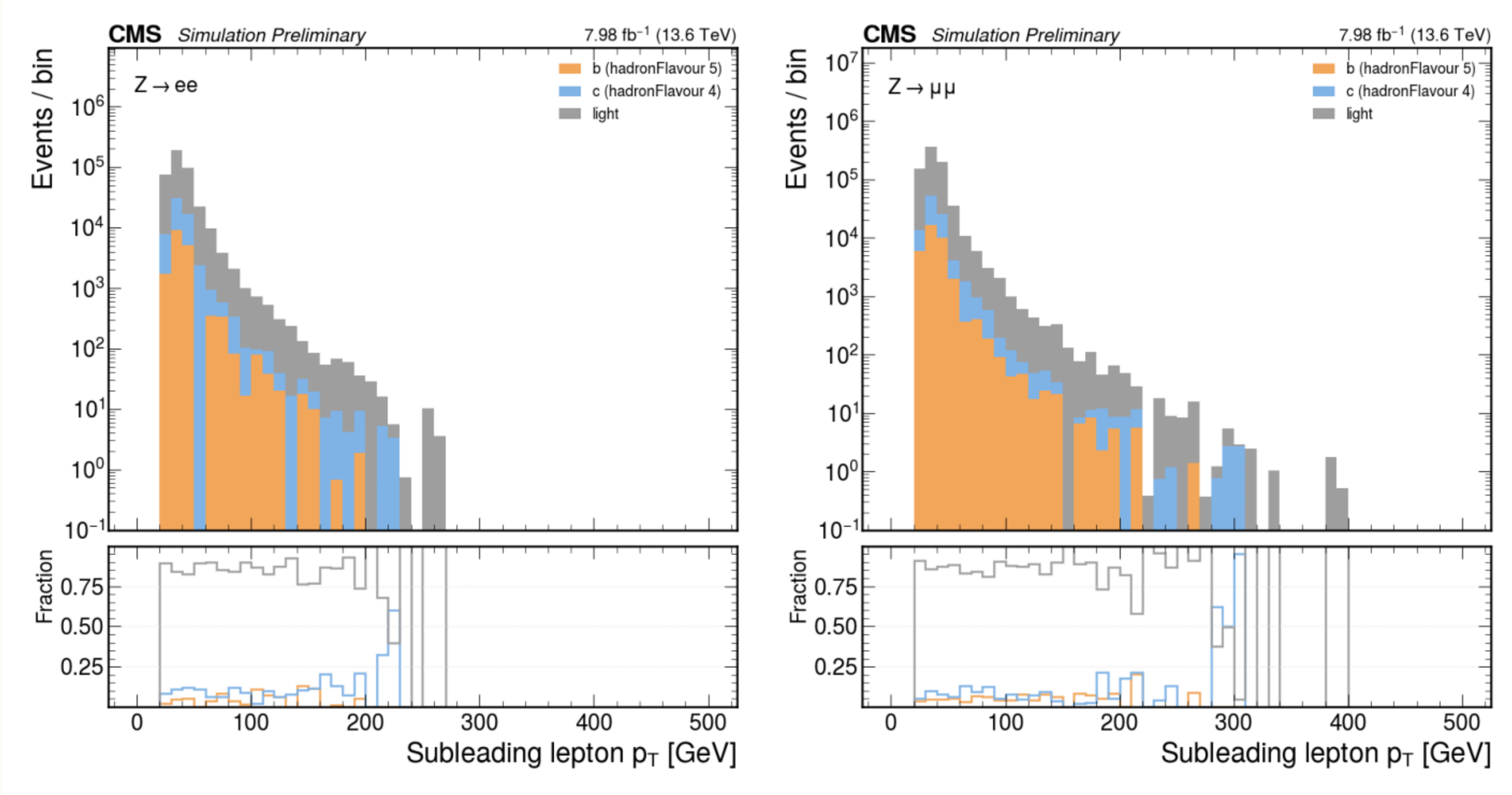
Leading lepton of the Z candidate, after the full selection (pT > 35 GeV required).




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 11

# Subleading-lepton pT

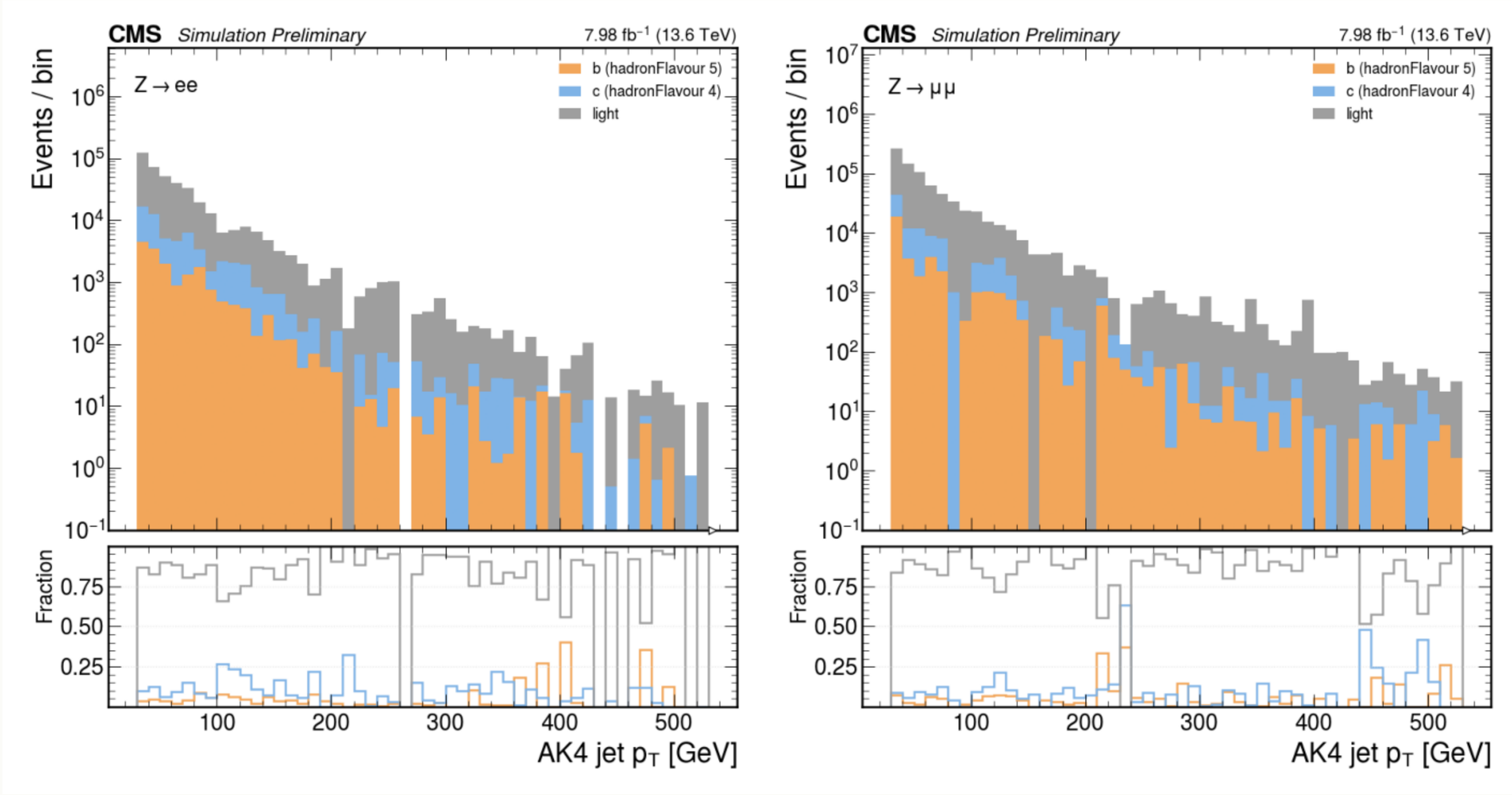
Subleading lepton of the Z candidate, after the full selection (pT > 25 GeV required).




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 12

# Leading AK4 jet pT

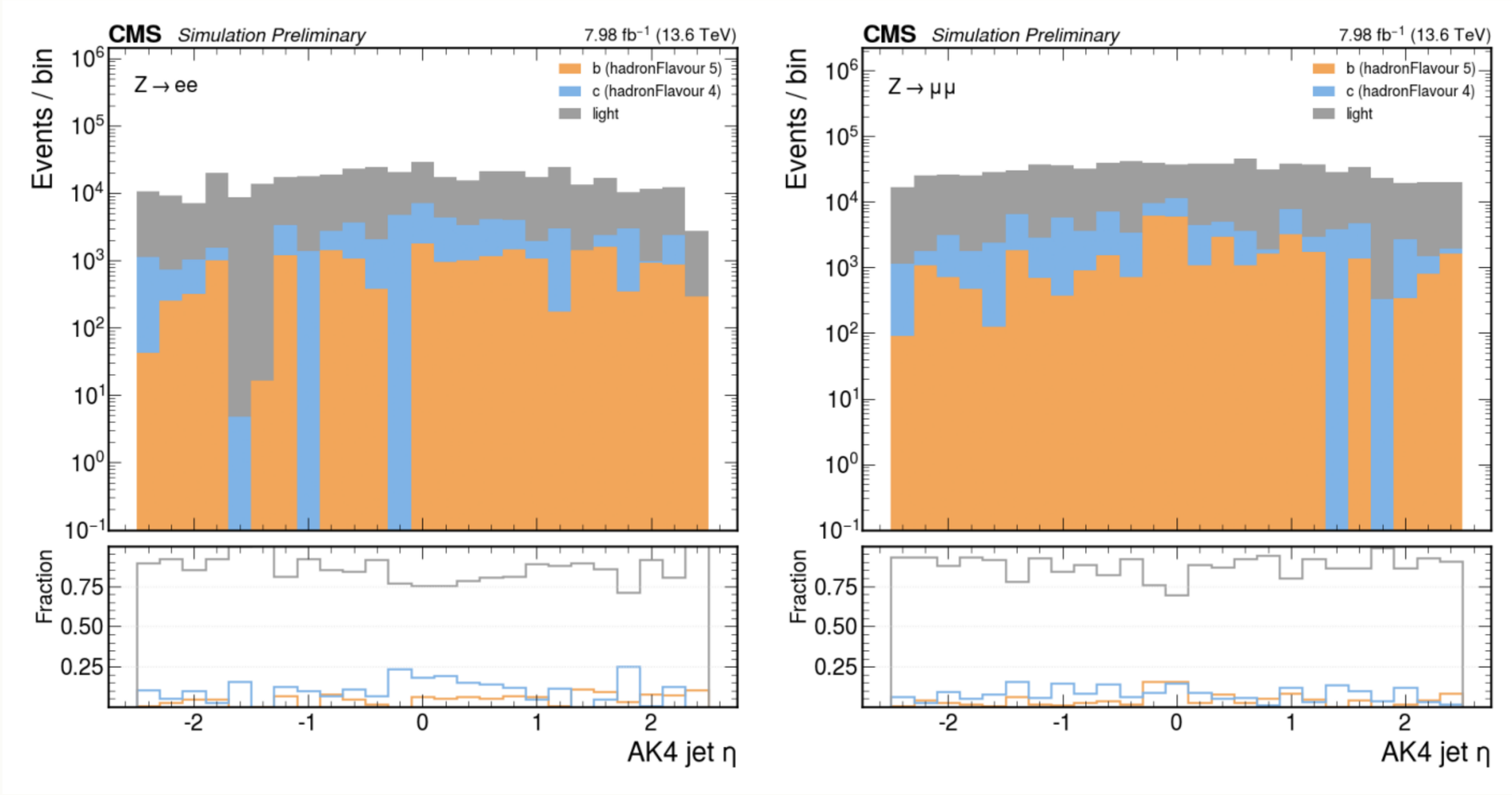
Leading lepton-cleaned AK4 jet with tight-lepton-veto ID (pT > 30 GeV required).




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 13

# Leading AK4 jet η

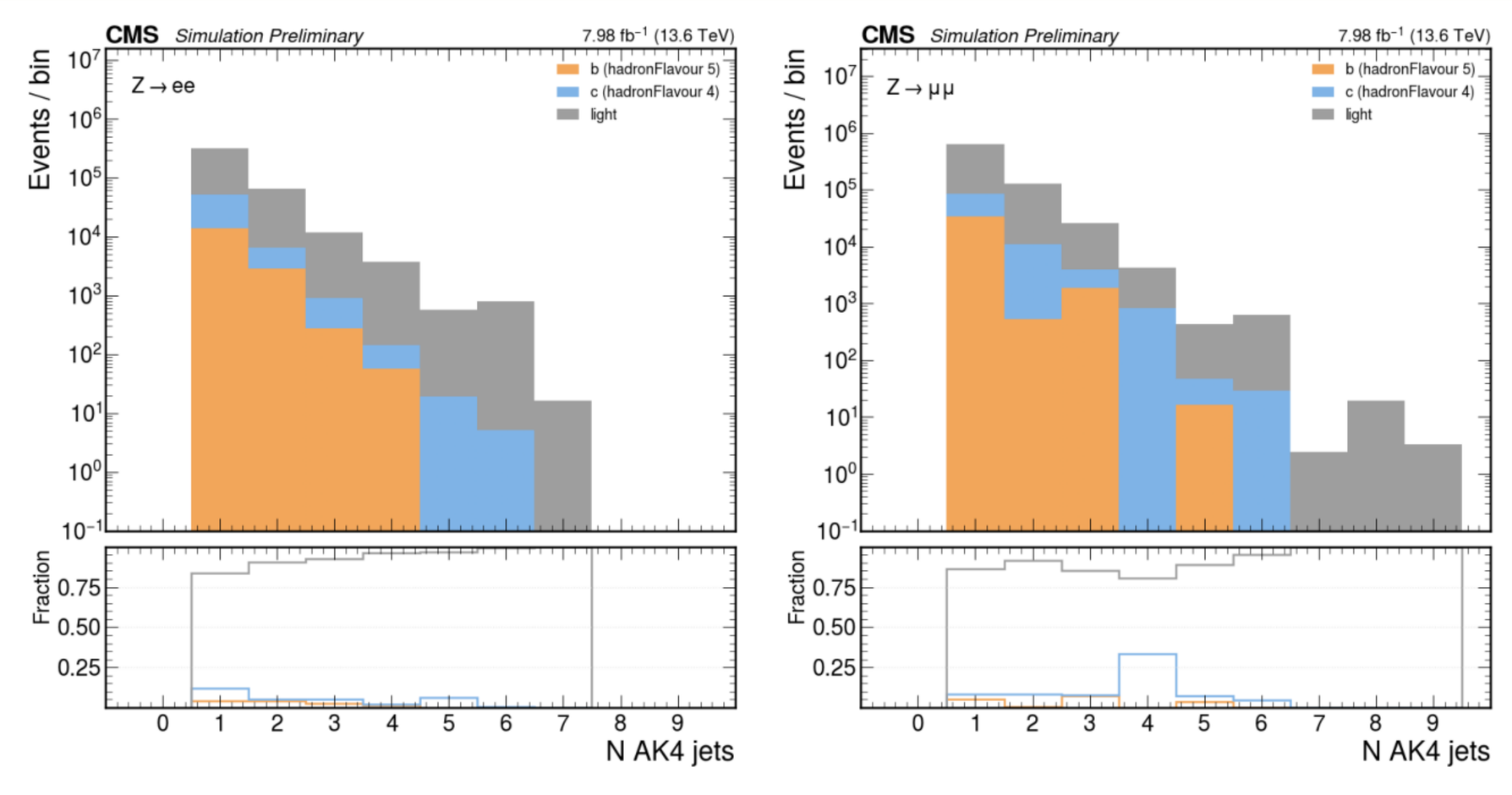
Pseudorapidity of the selected AK4 jet (|η| < 2.5 required).




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 14

# AK4 jet multiplicity

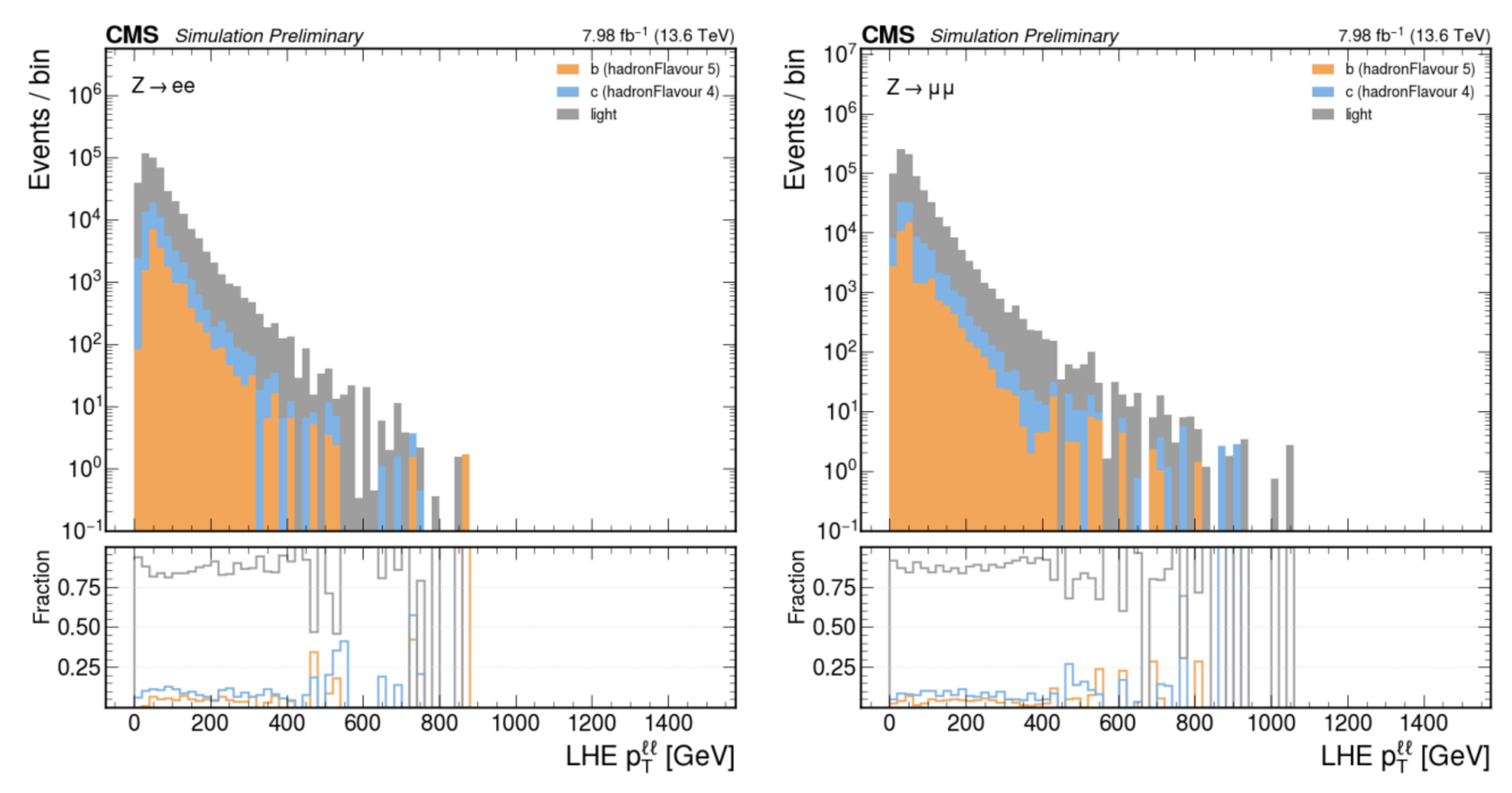
Number of selected AK4 jets (pT > 30 GeV, |η| < 2.5, ID, lepton-cleaned) in events passing the full selection.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 15

# LHE pT($\ell\ell$)

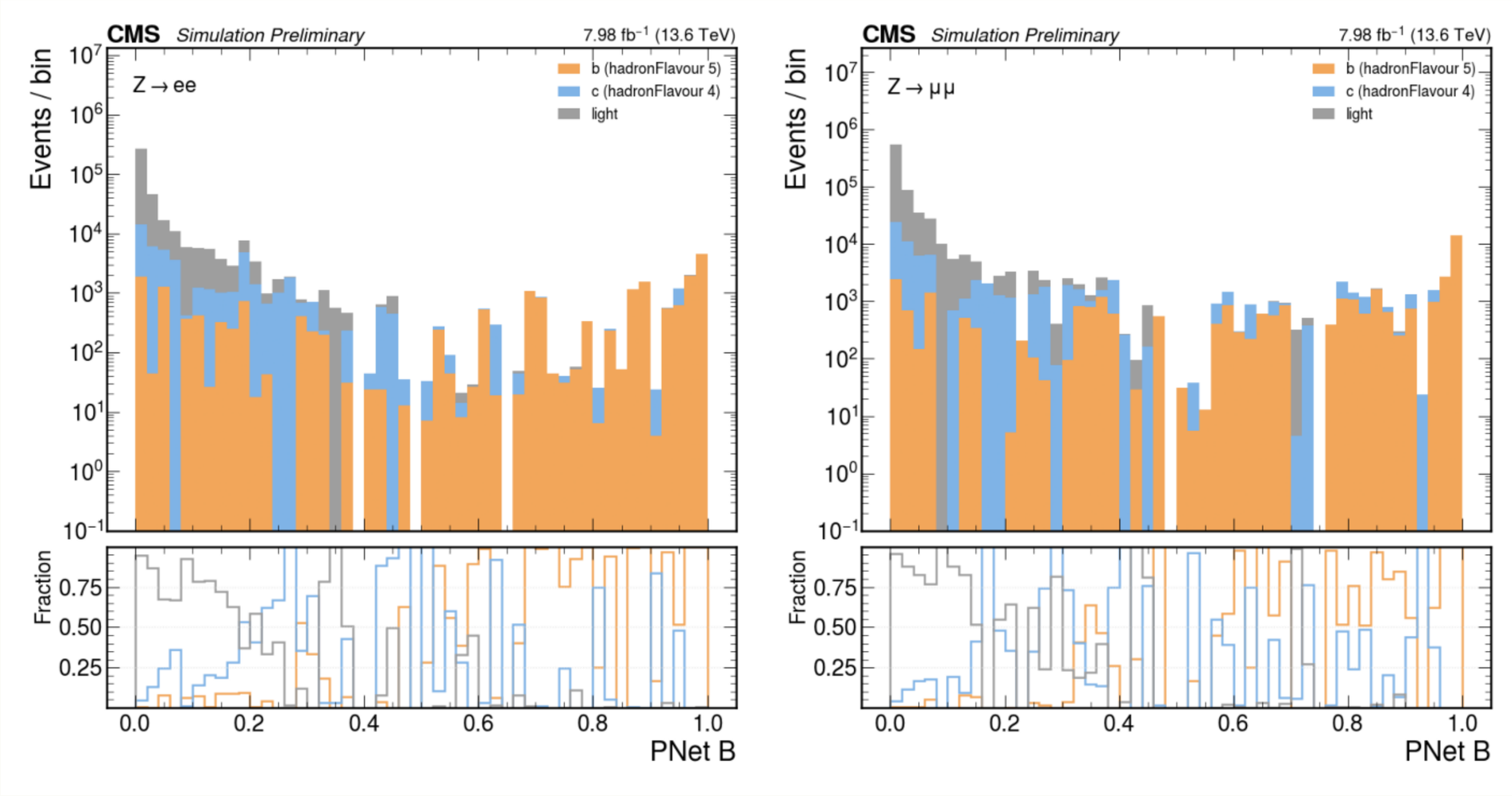
Generator-level pT($\ell\ell$) after the full selection: checks the sample stitching (slices at 100/200/400/600 GeV).




Summer22 DY→$\ell\ell$ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 16

# Leading AK4 jet: PNet B

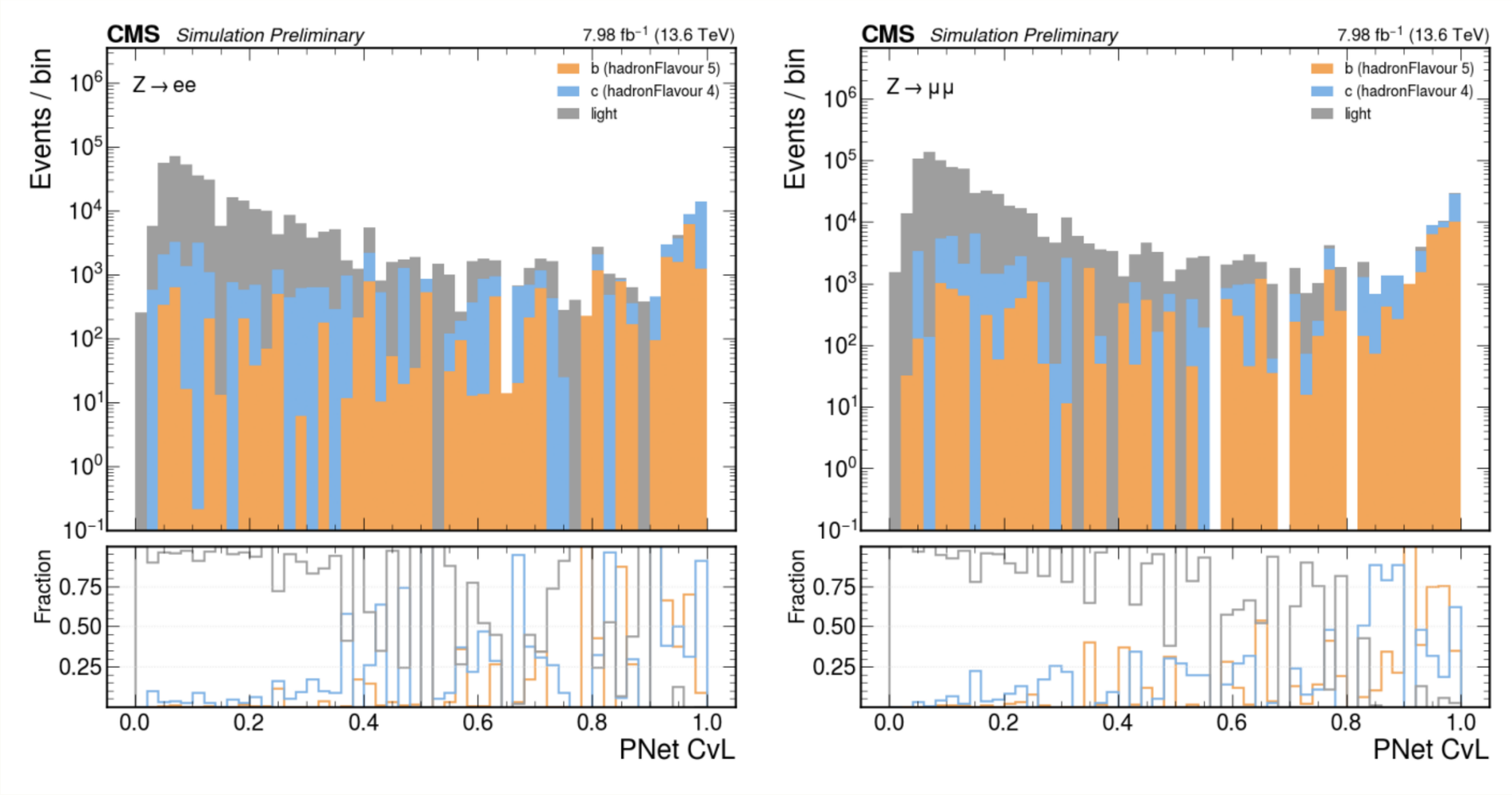
ParticleNet b-tag score of the selected jet; b jets pile up near 1, light jets near 0.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 17

# Leading AK4 jet: PNet CvL

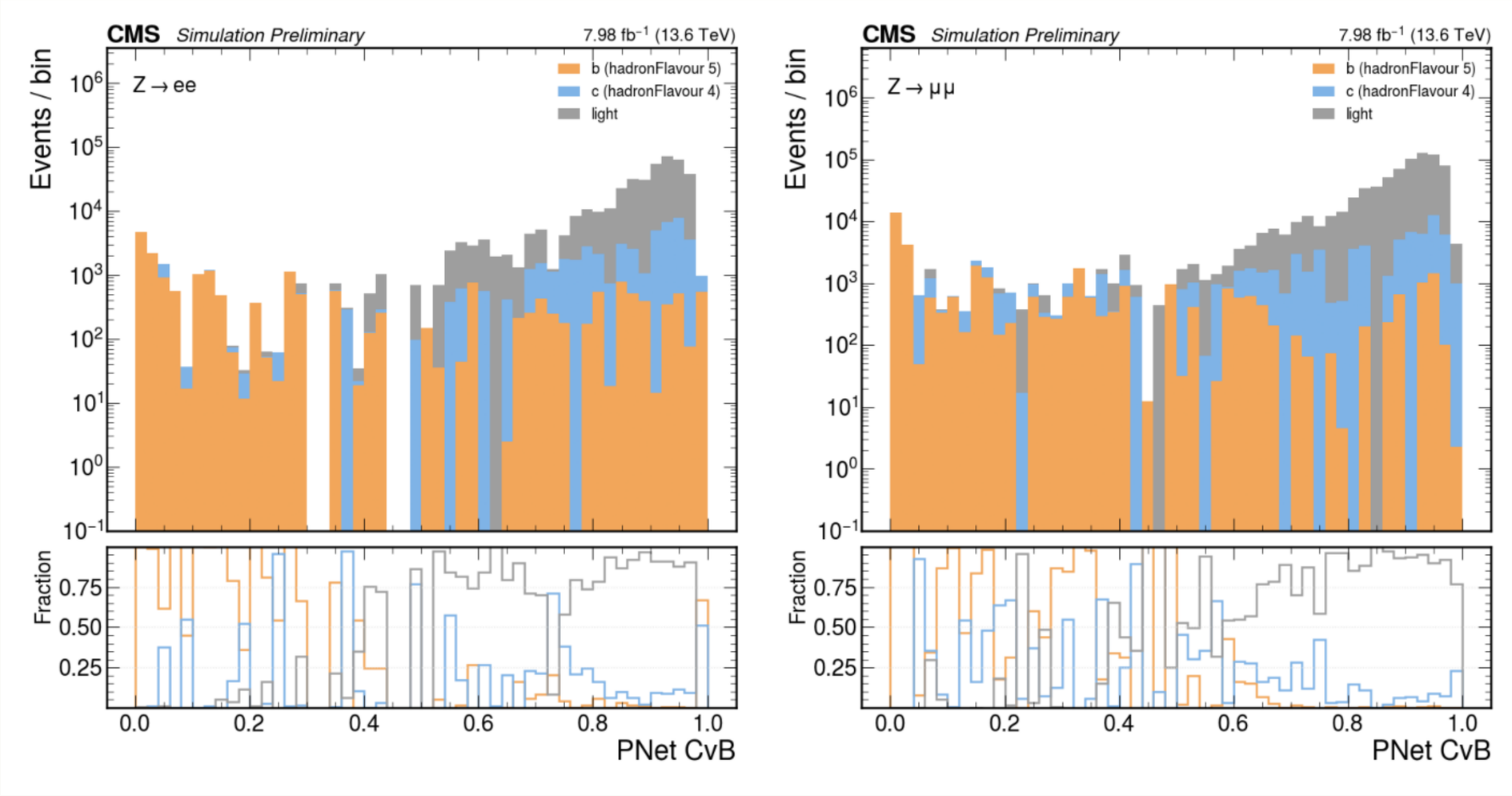
ParticleNet charm-vs-light score of the selected jet.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 18

# Leading AK4 jet: PNet CvB

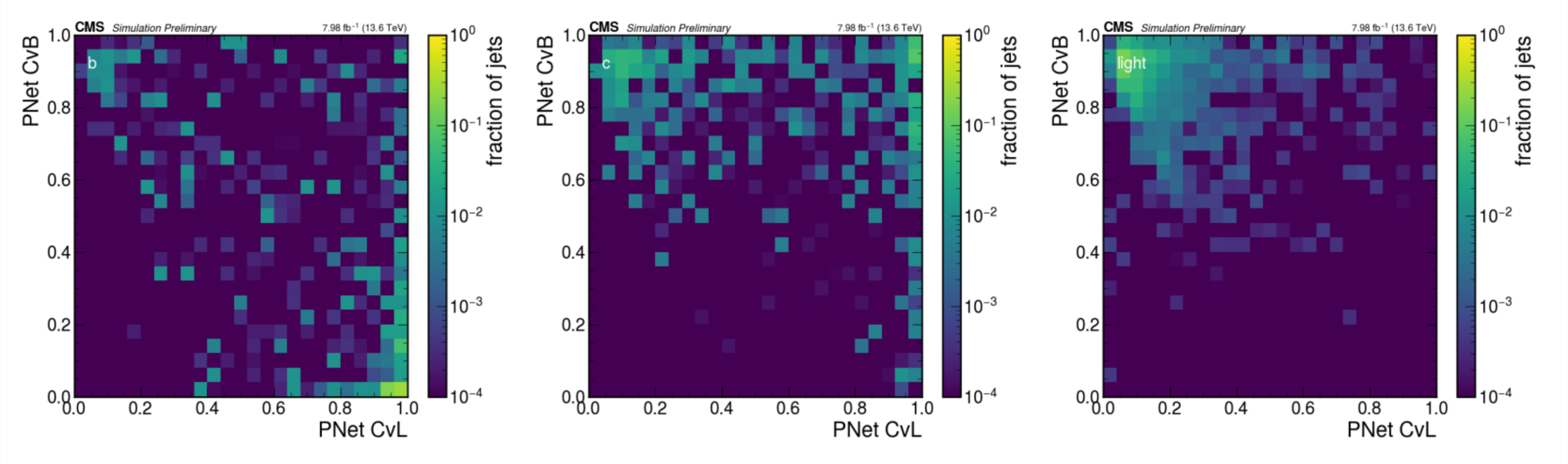
ParticleNet charm-vs-b score of the selected jet.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 19

# PNet charm-tagging plane: CvL vs CvB

Selected AK4 jets per flavour, both channels summed, normalized to unit area. b jets sit at high CvL and low CvB, light jets at low CvL and high CvB. c jets are at high CvB, but a large part of them overlaps the light-jet corner (low CvL), with the rest at high CvL.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · Z→ee + Z→μμ · 20

# Jet flavour: what CMS recommends

## CMS recommendation (BTV)

- Use the hadron-based flavour (hadronFlavour), the CMSSW default
- Generator B and C hadrons are clustered into the jet with ~zero momentum, so the jet itself is unchanged
- b if the jet has a B hadron, c if it has a C hadron but no B, light otherwise
- All BTV b/c-tag scale factors are given for this definition
- partonFlavour: mainly to separate gluon from uds jets

## What we do

- Take Jet_hadronFlavour from NanoAOD for the leading jet: 5 = b, 4 = c, rest = light
- No matching of our own
- No b/bb or c/cc split: NanoAOD doesn't store hadron counts for AK4 jets
- partonFlavour only as a cross-check: agrees for 99% of b jets and 92% of c jets

## Conclusion: we are on the right track

Our flavour labels are exactly the ones CMS recommends, so the BTV scale factors can be applied as they are. The c/b ratio in our MC (2.5 for Z→ee, 1.8 for Z→μμ) is a bit higher than, but in the same ballpark as, the CMS 13 TeV measurement, 1.62 ± 0.15 (PRD 102, 032007). Not a like-for-like comparison yet: ours is MC truth for the leading jet, without tagging.

**Reference: CMS TWiki, Jet Flavour Identification (MC Truth)**

https://twiki.cern.ch/twiki/bin/view/CMSPublic/SWGuideBTagMCTools#Hadron_parton_based_jet_flavour

# hadronFlavour vs partonFlavour

## hadronFlavour (default)

- Ghost-matched B / C hadrons clustered into the jet
- b (5) if any B hadron, else c (4) if any C hadron, else 0
- BTV standard; b/c-tag SFs are measured for it

## partonFlavour

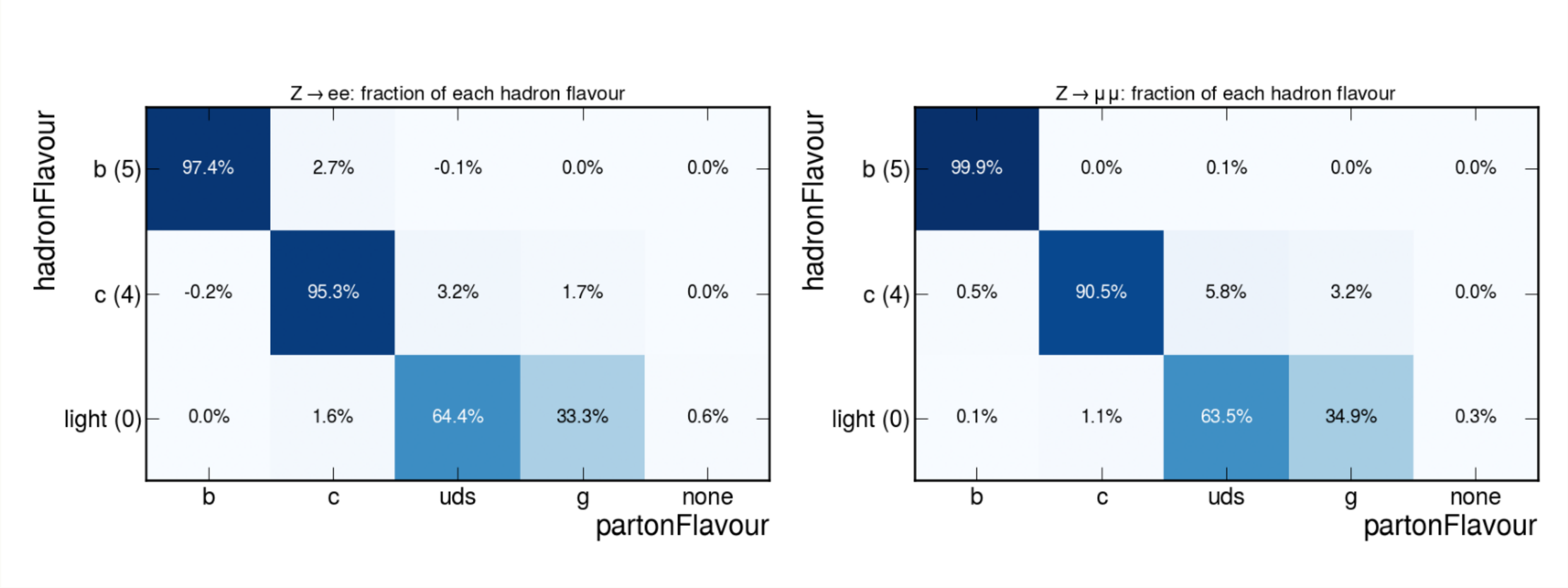
- Ghost-matched partons (last before hadronization) in the jet
- Hardest heavy parton wins: b (5), c (4), else hardest light parton
- Splits light jets into uds (1–3) and gluon (21); 0 = no parton matched

| Fraction of scaled yield | hadron Z→ee | parton Z→ee | hadron Z→μμ | parton Z→μμ |
|---|---|---|---|---|
| b | 0.042 | 0.041 | 0.045 | 0.047 |
| c | 0.106 | 0.116 | 0.083 | 0.085 |
| light (hadron) / uds + g + none (parton) | 0.852 | 0.843 | 0.872 | 0.868 |
| of which g / none (parton) | | 0.286 / 0.005 | | 0.307 / 0.003 |

Summer22 DY→$\ell\ell$ MC · 1 file per sample · AK4 jets · leading AK4 jet, scaled to 7.98 fb$^{-1}$ · 22

# Hadron vs parton flavour migration

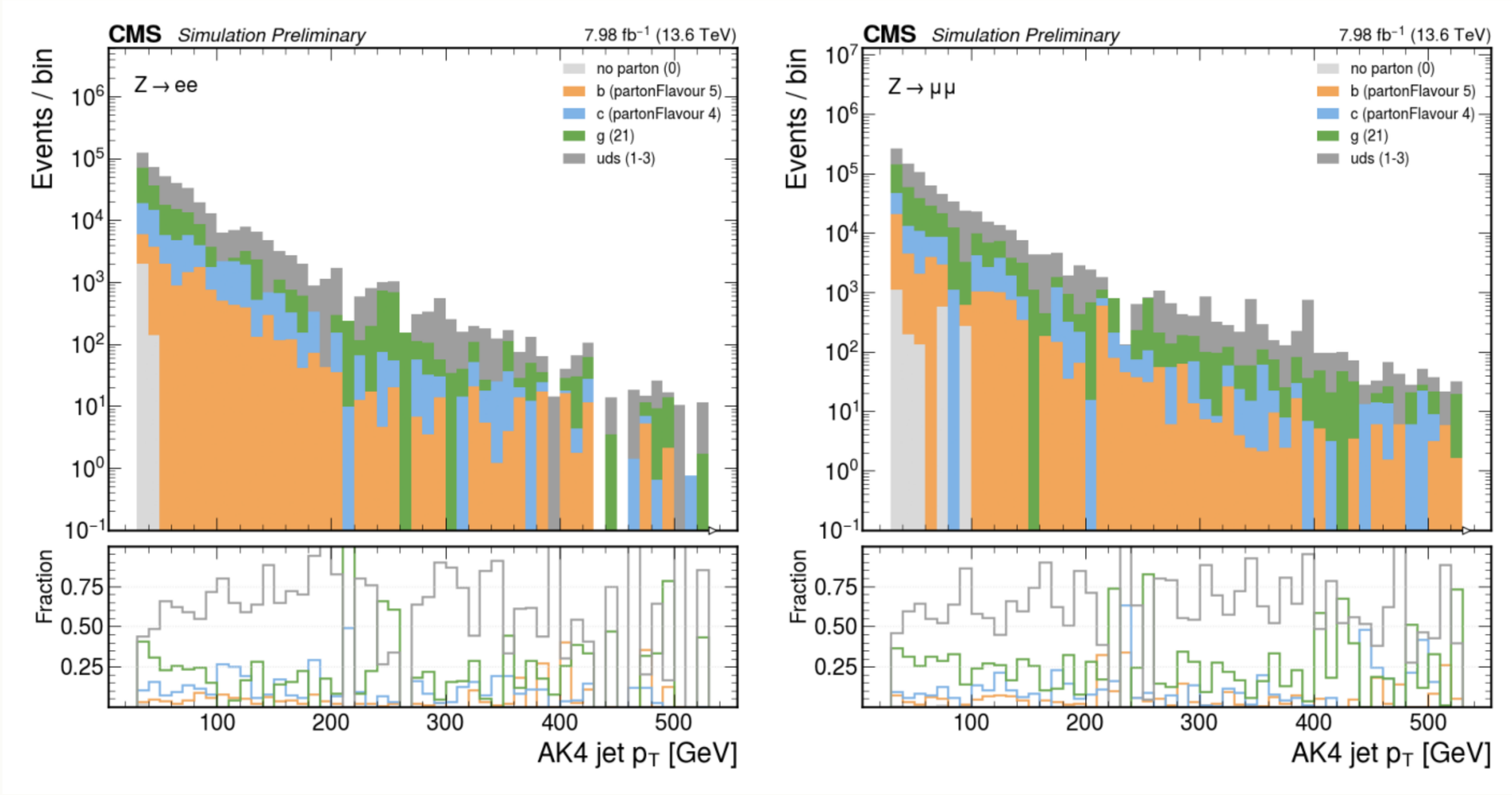
Each row shows where the jets of one hadronFlavour land in partonFlavour. 99% of hadron-b and 92% of hadron-c jets also have partonFlavour b / c (both channels). Hadron-light jets split into uds (64%) and gluon (34%); 1.4% carry a c or b parton without a matching hadron.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · rows: hadronFlavour, each sums to 100% (scaled) · small negative entries = NLO negative weigh

# Leading AK4 jet pT, by partonFlavour

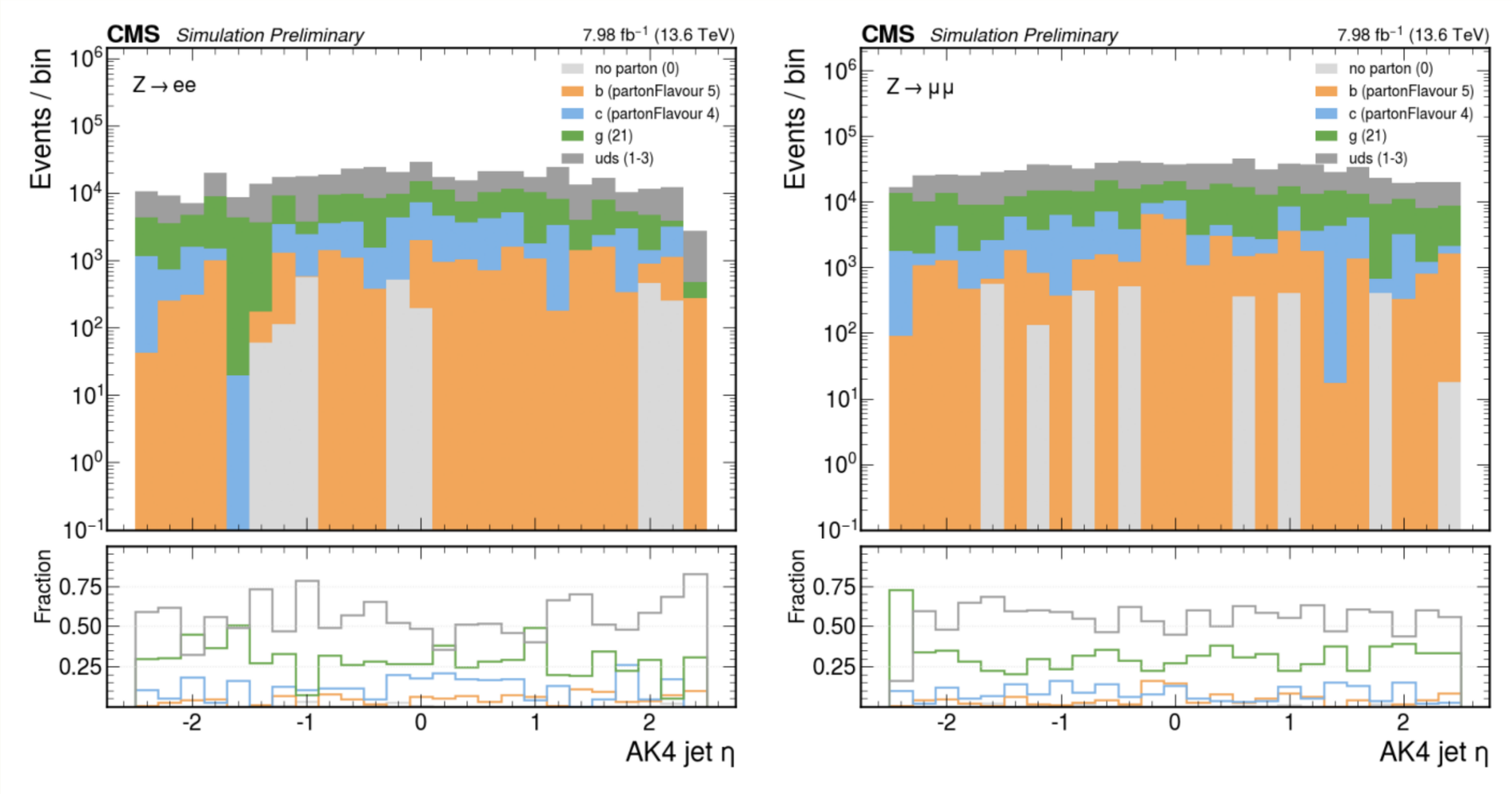
Same as page 13, stacked by |partonFlavour| of the leading jet.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · by leading-jet partonFlavour; lower: fraction per bin · Z→ee left, Z→μμ right · 24

# Leading AK4 jet η, by partonFlavour

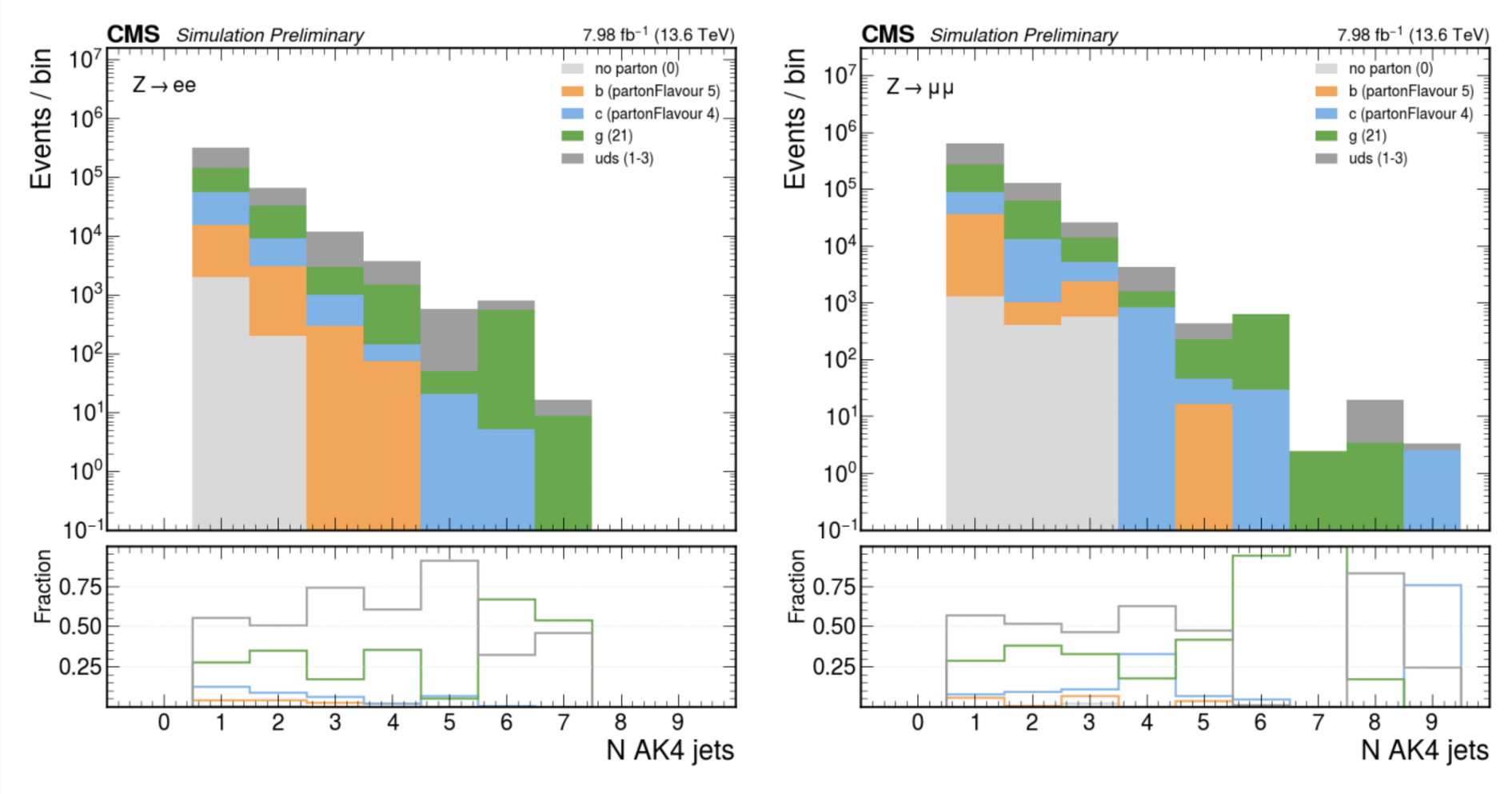
Same as page 14, stacked by |partonFlavour| of the leading jet.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · by leading-jet partonFlavour; lower: fraction per bin · Z→ee left, Z→μμ right · 25

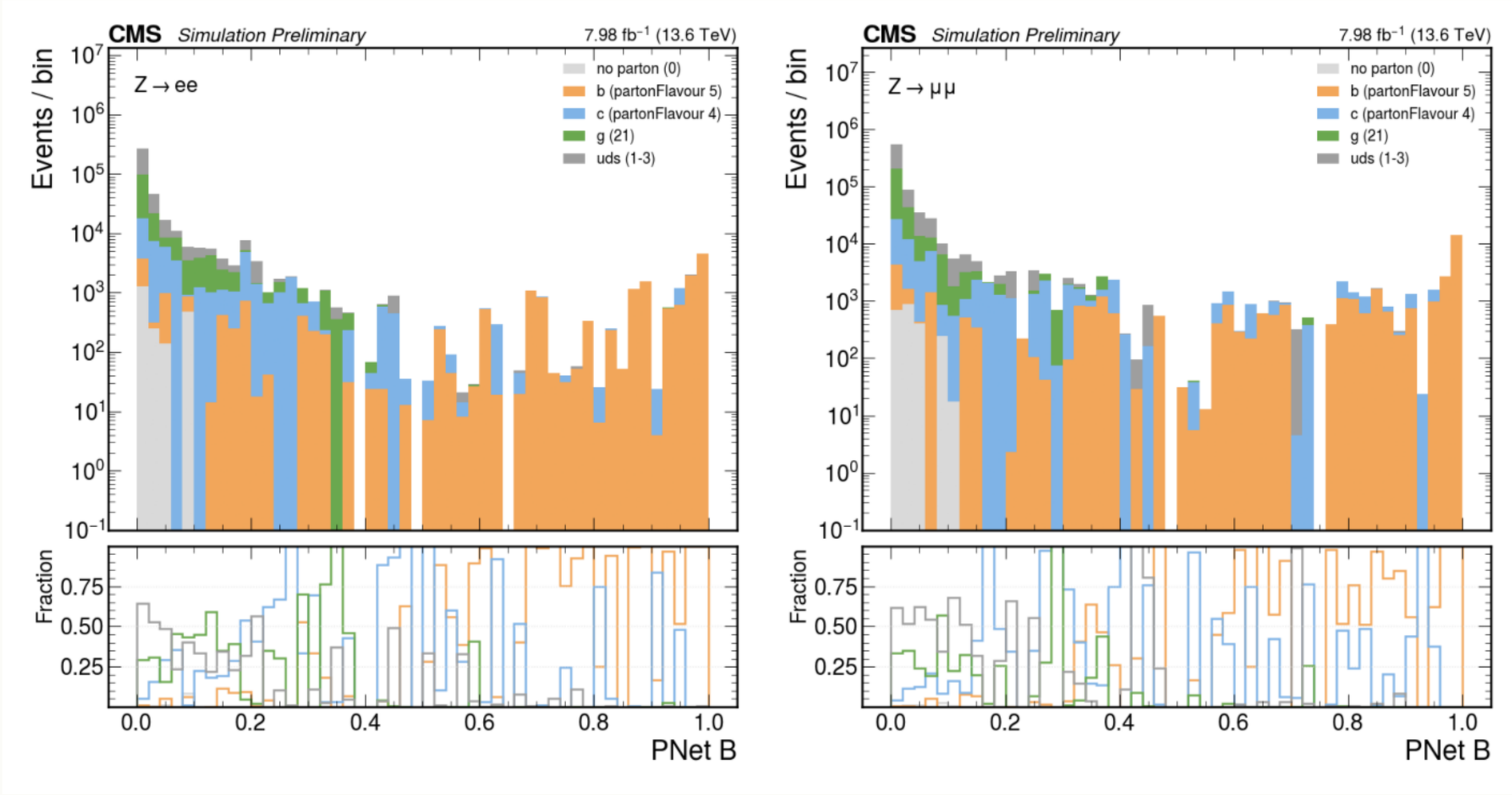
# AK4 jet multiplicity, by partonFlavour

Same as page 15, stacked by |partonFlavour| of the leading jet.



# PNet B, by partonFlavour

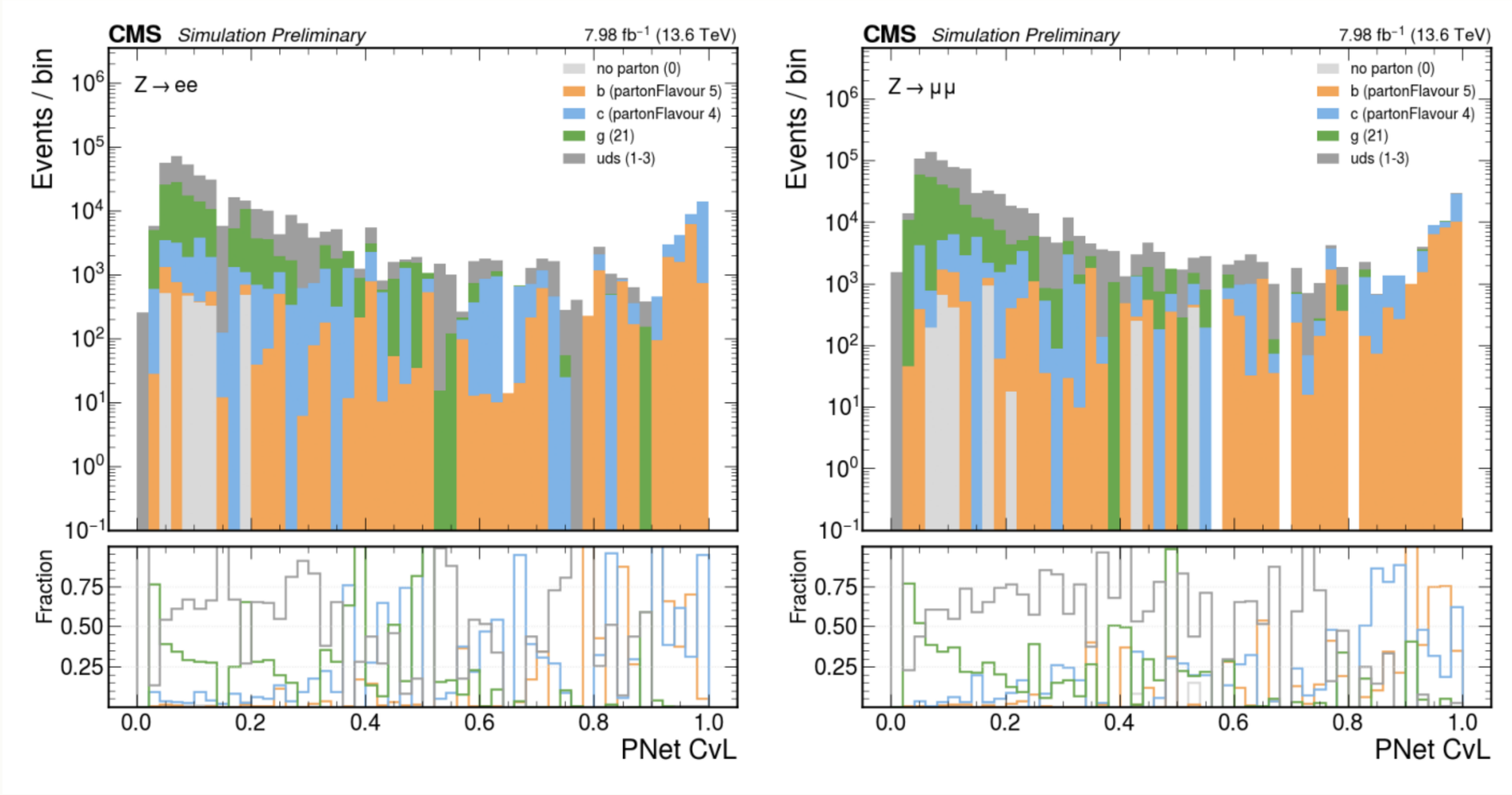
Same as page 17, stacked by |partonFlavour| of the leading jet.




Summer22 DY→ℓℓ MC · 1 file per sample · AK4 jets · by leading-jet partonFlavour; lower: fraction per bin · Z→ee left, Z→μμ right · 27

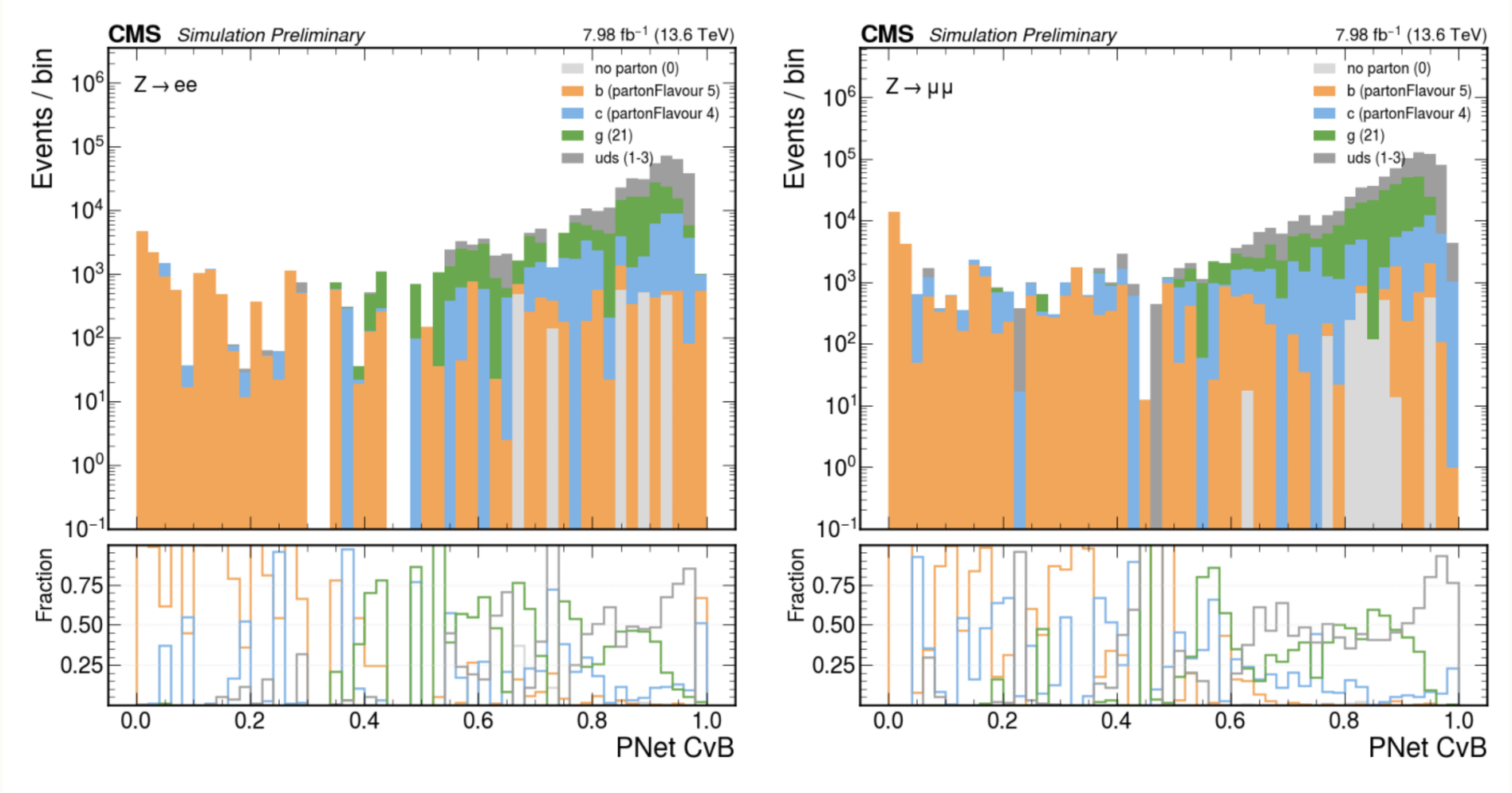
# PNet CvL, by partonFlavour

Same as page 18, stacked by |partonFlavour| of the leading jet.



# PNet CvB, by partonFlavour

Same as page 19, stacked by |partonFlavour| of the leading jet.



# No-parton jets are pileup

## What we saw

In 0.4% of events the leading jet has no parton attached (partonFlavour 0). They are all light jets, and they are soft: 78% are below 50 GeV, compared with 51% for all leading jets.

## How we checked

A jet from the main collision should have a generator-level jet (GenJet) next to it. NanoAOD stores the match as Jet_genJetIdx, which is −1 when there is none within $\Delta R < 0.4$. 100% of the no-parton jets have no GenJet, against 0.03% for b, c and uds jets. No generator partner means the jet comes from another collision in the same bunch crossing: pileup.

## Why we keep them

We can't use generator information in data, so we can't remove these jets by truth in MC either. Data has pileup jets too, and the MC has to describe them. Since they are light jets and a tiny fraction of the yield, Z+b and Z+c are not affected.

### Bottom line: expected, small, and harmless for the b/c ratio

No action needed. Our AK4 jets are PUPPI jets, which already remove most pileup. If we ever want fewer, the handle is a reco-level cut (e.g. a higher jet pT threshold), applied the same way to data and MC.

# Leading jets with partonFlavour 0 are pileup

A jet with no GenJet (pT > 10 GeV) within ΔR < 0.4 (Jet_genJetIdx = −1) is a pileup jet. All 20 leading jets with partonFlavour 0 have no GenJet match (8 Z→ee, 12 Z→μμ).

| Leading-jet partonFlavour | Z→ee no GenJet / all | Z→ee % | Z→μμ no GenJet / all | Z→μμ % |
|---|---|---|---|---|
| b | 0 / 1,203 | 0.00 | 0 / 1,714 | 0.00 |
| c | 1 / 1,980 | 0.05 | 2 / 3,026 | 0.07 |
| uds | 7 / 12,991 | 0.05 | 4 / 20,170 | 0.02 |
| g | 20 / 5,185 | 0.39 | 37 / 8,184 | 0.45 |
| none | 8 / 8 | 100.00 | 12 / 12 | 100.00 |
| all | 36 / 21,367 | 0.17 | 55 / 33,106 | 0.17 |

- Not every pileup jet has partonFlavour 0: most unmatched leading jets carry a gluon label (a soft ISR/UE gluon is ghost-matched, but there is no GenJet).
- Pileup barely reaches the heavy-flavour yields: 0 of 2,842 hadron-b and 1 of 4,753 hadron-c leading jets are unmatched. Scaled, 1.9% (Z→ee) and 1.3% (Z→μμ) of leading jets are unmatched. This is above the raw fraction because the events come from the high-weight inclusive sample, so it waits for the full MC run.

# N_jet bumps go away with more MC

20 files per sample instead of 1. Z→ee N_jet = 6: 801.9 (N_eff 1.8) → 45.5 (N_eff 0.7); Z→μμ N_jet = 6: 630.0 (N_eff 1.1) → 108.4 (N_eff 2.9), N_jet = 8: 19.7 (N_eff 1.4) → 0.0 (N_eff 0.0). Both spectra now fall smoothly up to N_jet = 7. The new Z→ee N_jet = 8 point, 33.7 (N_eff 1.1), is again a single inclusive-sample event: N_jet ≥ 6 still needs the full run.

